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Copper binding and reactivity at the histidine brace motif: insights from mutational analysis of the *Pseudomonas fluorescens* copper chaperone CopC

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The histidine brace (His-brace) is a copper-binding motif that is associated with both oxidative enzymes and proteinaceous copper chaperones. Here, we used biochemical and structural methods to characterize mutants of a His-brace-containing copper chaperone from *Pseudomonas fluorescens* (PfCopC). A total of 15 amino acid variants in primary and second-sphere residues were produced and characterized in terms of their copper binding and redox properties. PfCopC has a very high affinity for Cu(II) and also binds Cu(I). A high reorganization barrier likely prevents redox cycling and, thus, catalysis. In contrast, mutations in the conserved second-sphere Glu27 enable slow oxidation of ascorbate. The crystal structure of the variant E27A confirmed copper binding at the His-brace. Unexpectedly, Asp83 at the equatorial position was shown to be indispensable for Cu(II) binding in the His-brace of PfCopC. A PfCopC mutant that was designed to mimic the His-brace from lytic polysaccharide monooxygenase-like family X325 did not bind Cu(II), but was still able to bind Cu(I). These results highlight the importance of the proteinaceous environment around the copper His-brace for reactivity and, thus, the difference between enzyme and chaperone.

Keywords: ascorbate assay; isothermal calorimetry; protein; X-ray crystal structure

Copper enzymes are involved in a range of biological processes such as electron transfer, oxygen activation and nitrite metabolism [1,2]. This diverse functionality can be attributed to the geometry and electronic structure of copper. Cu(I) adopts a closed shell, $3d^{10}$ electronic configuration, and prefers a symmetric tetrahedral coordination geometry, while Cu(II) has a $3d^9$ electron configuration with degenerate orbitals that leads to an energy-lowering Jahn-Teller distortion. This distortion generally leads to elongation of the two axial bonds and contraction of the four equatorial bonds resulting in a limiting square planar geometry [1]. Enzymes that use copper as co-factor in redox reactions must thus accommodate these two very

Abbreviations
BCA, bicinchoninic acid; DSF, differential scanning fluorimetry; ITC, isothermal titration calorimetry; LPMO, lytic polysaccharide monooxygenase.

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different geometries. In proteins, the copper-coordinating atoms can be nitrogen, oxygen or sulphur and the coordination geometry is usually well defined in a discrete copper coordination site [3].

The histidine brace (His-brace) motif is an example of copper coordination. This motif consists of an N-terminal histidine and an internal histidine that donate three nitrogen atoms for chelation. Key features of this motif are that two of the nitrogen atoms are derived from the imidazole side chains while the final nitrogen atom is donated by the N terminus of the protein. This motif is known from the lytic polysaccharide mono-oxygenases (LPMOs), where it is thought to enable copper reactivity [4–6]. The His-brace is also found in the particulate methane mono-oxygenases but some controversy remains as to whether the active site can indeed be attributed to the His-brace [7,8]. Furthermore, this motif is also found in the large family of bacterial copper-binding proteins CopC [9,10]. CopC is part of the *cop* system that affords these bacteria the ability to control copper availability [10,11]. The CopC family can be divided into Type A (two copper-binding sites) and Type B (one copper-binding site). An example of a Type B CopC is found in *Pseudomonas fluorescens* (PfCopC). Wijekoon *et al.* [12] determined that PfCopC had a subfemtomolar affinity for Cu(II), using competing fluorescent probes. Using site-directed mutations, the residues His1 and His83 have been found to be most important for the extremely high copper affinity of PfCopC. Although His3 was also believed to be important, because it is part of a putative NH_2X-X-His3 amino terminal copper- and nickel-binding motif (ATCUN). It was later shown by X-ray crystallography [13], and biochemical assays [14], that native PfCopC coordinates copper in a His-brace formed by His1 and His83 with an additional carboxylic group from the side chain of Asp83. His1 and His83 coordinate copper through the δN of their imidazole side chains. Two other residues have also been proposed to be important for the high copper affinity, namely the strictly conserved Glu27 and the partially conserved Ala2 [14,15].

A permutation of the PfCopC copper His-brace has been found in the X325 LPMO-like family, where no redox activity has been observed [14,16,17]. The X325 His-brace has the two imidazole side chains in the *trans*-position. As in LPMOs, His1 coordinates through the δN and the internal histidine coordinates through the εnitrogen [18]. Redox chemistry of model His-Gly-His peptides has been shown to be very dependent on the tautomer used for coordination [19]. Only Cu(I)-peptides that employ the εN for coordination were shown to be reactive towards oxygen.

Another biologically relevant copper His-brace has been found in the viral spindles of the crystalized fusolin protein. Fusolin is very likely an LPMO, as it has structural similarities to chitin-active LPMOs. These spindles could be inactive forms of LPMOs, where an Asp from a neighbouring protein molecule in the natural spindle crystal caps the His-brace [20]. We have used PfCopC as a model system to study copper chelation and reactivity at the His-brace. In particular, we investigated whether the removal of Asp83 from the copper coordination at the His-brace would enable redox cycling, and thus transform CopC from a chelator into a potential redox enzyme.

Methods

All chemicals were obtained from Merck Sigma-Aldrich (Darmstadt, Germany) unless otherwise stated and were of reagent grade or better.

Protein expression and purification

The mutants were generated from a synthetic gene of PfCopC (NCBI WP043048726) using uracil excision cloning, as described previously [21]. Constructs were generated in a pET28a-MalE^{SP} plasmid and transformed into BL21 (DE3)-competent cells for expression and secretion into the periplasm. Cells were grown in LB medium supplemented with kanamycin at 30 °C and 250 r.p.m. Expression was induced at mid-late exponential phase with 1 mM IPTG and cells were harvested 20 h after induction. The periplasmic fraction was isolated as described previously [22], and filtered using a 0.22 µm filter (Frisenette, Knebel, Denmark). The periplasmic fraction was diluted in 20 mM Tris buffer at pH 8.3, and loaded onto a Q Sepharose column (GE Healthcare, Chicago, IL, USA). Tris buffer was made by mixing Trizma and Tris–HCl in appropriate amounts to give the desired ionic strength and pH. Under these conditions, CopC does not bind to the column and elutes in the flow-through. The flow-through was concentrated using a continuously chilled Amicon-stirred cell system operated in concentration mode. A 5 kDa MWCO PES filter (Merck) was used as the size filter. The concentrate was loaded onto a 16/600 Superdex-75 pg column equilibrated with 40 mM MES with 150 mM NaCl at pH 6.6 (adjusted with HCl). Protein eluted as a monodisperse peak, as confirmed by SDS/PAGE Amino acid compositional analysis was used to identify and determine the concentrations of amino acids in the final fraction [23].

Ascorbate oxidation assay

The ascorbate oxidation assay described by Yako *et al.* [24] was adapted as follows. Solutions were prepared in 20 mM MES buffer at pH 6.6. To remove metal impurities buffers were extensively treated with Chelex© 100 according to manufacturer's instructions prior to experiments. Experiments were performed in a BioTek Synergy H1 plate reader (Agilent, Santa Clara, CA, USA). PfCopC mutants were loaded overnight at 4 °C in a stoichiometric range from 1 : 0.25 to 1 : 1. The next day the assay was initiated by the addition of freshly prepared ascorbate to a final concentration of 100 µM, and the change in absorbance at 265 nm was read once a minute for 150 min. Technical triplicate experiments were performed. For scavenger experiments, the assay was performed in the presence of either 20 µM BCA or 1 nM catalase. These experiments were also performed in triplicate. H_2O_2 generation was monitored with an Amplex® UltraRed kit (Thermo Fischer, Waltham, MA, USA). Assay preparation was carried out in the same way as for the ascorbate assay. PfCopC (4 µM) and mutants were equilibrated with $CuCl_2$ (2 µM) overnight. The ascorbate assay was performed in the presence of 0.1 $U{\cdot}L^{-1}$ horseradish peroxidase and 50 µM Amplex UltraRed reagent. The change in fluorescence was monitored at 590 nm using excitation at 530 nm.

Isothermal titration calorimetry

Experiments were carried out on a MicroCal PEAQ-ITC instrument (Malvern Panalytical, Malvern, UK). Data were obtained using a 200 µL sample cell and a 40 µL titration syringe with an injection volume of 3 µL. The resulting data were fitted to a single-site binding model to determine the thermodynamic parameters and stoichiometry. For anaerobic experiments, the instrument was placed in a custom-made rigid acrylic glovebox (Belle Technology UK Ltd, Charlestown, UK). Prior to use, the glovebox was purged with N_2 and the atmosphere maintained at < 0.8 p.p.m. O_2 using a stand-alone recirculation unit equipped with molecular sieve cartridges. Anaerobic copper solutions were prepared as described previously [25], using CuCl as the source of Cu(I). The solutions were prepared under anaerobic conditions in the presence of acetonitrile (MeCN). All buffers were thoroughly purged with N_2 for 2 h with stirring in infusion flasks. Degassed buffers were kept in the glovebox for the duration of the experiments. PfCopC mutant samples were overlaid with argon gas for 2 min and placed in the glovebox. Inside the glovebox, the protein samples were stirred for 1.5 h at 300 r.p.m. before the experiments. MeCN was used to stabilize Cu(I) to prevent oxidation to Cu(II). The Cu(II) content in copper solutions was measured by isothermal titration calorimetry (ITC) using EDTA as the titrant. Similarly, the $[Cu(I)(MeCN_4)]^+$ content was measured using BCS as the titrant.

Turnover frequency

The turnover frequency was determined as the linear rate at the beginning of the progress curve, v_0, normalized to the concentration of enzyme and the absorbance signal range. The full signal range of oxidation of 100 µM ascorbate was determined from the difference between samples without copper and samples with 0.5 µM $CuCl_2$. Concentrations of H_2O_2 were determined from a standard curve for Amplex® UltraRed according to the manufacturer's instructions. The turnover frequency was defined as:

$$v = \frac{v_0}{c} \cdot \frac{100\ \mu\text{M}}{\Delta}.$$

UV-Vis spectroscopy

Samples of 450 µM loaded with one equivalent of $CuCl_2$ were prepared. Absorbance spectra were recorded in 1 cm quartz cuvettes with septum (FireflySci, New York, NY, USA) using a Perkin Elmer Lambda 800 spectrophotometer (Waltham, MA, USA). Samples from reduction experiments were mixed with ascorbic acid to a final concentration of 2 mM. Spectra were recorded at 5-min intervals over a 2-h period. Samples from anaerobic reduction experiments were overlaid with argon and placed inside the glove box for stirring as described for the samples for ITC experiments. The cuvettes were open only inside the anaerobic environment.

Differential scanning fluorimetry

The inflection points for thermal unfolding of the PfCopC and its mutants were determined using a Tycho NT.6 nano (München, Germany) differential scanning fluorimetry instrument, which monitors intrinsic fluorescence from the aromatic residues of the protein, with a temperature ramp of 30 $°C{\cdot}min^{-1}$, applied from 35 °C to 95 °C. Stock solutions were prepared using protein concentrations of 0.5 $mg{\cdot}mL^{-1}$ in 40 mM MES at pH 6.6 with 150 mM NaCl without copper acetate and with copper acetate at a 1.2 molar ratio to the protein. All experiments were performed in technical triplicates using the same stock solution.

Crystallization

Crystallization of the PfCopC D83A, D83N and E27A mutants was accomplished using the sitting-drop vapour diffusion method. Drops with a volume of 0.3 µL were set up by an Oryx8 robot (Douglas Instruments, East Garston, UK) in MRC 2 drop plates (Molecular Dimensions). All the proteins were kept in 40 mM MES at pH 6.6, with 150 mM NaCl, and were initially screened using the commercially available screens, JCSG+, Morpheus (Molecular Dimensions, Sheffield, UK), Pact (Qiagen, Hilden, Germany) and SaltRX (Hampton Research, Aliso Viejo, CA, USA). Prior to crystallization, the proteins were incubated for 30 min with equimolar copper acetate. Optimization of SaltRX condition F4 gave fully grown PfCopC D83A and D83N crystals in 4 weeks at room temperature, in drops consisting of 75% protein solution and 25% reservoir (2.5 M ammonium sulphate, 0.1 M sodium acetate at pH 4.6 for D83A; 2.9 M ammonium sulphate and 0.1 M sodium acetate at pH 4.6 for D83N). Optimization and additive screening (Hampton Research kit) gave a PfCopC E27A crystal in 3 weeks at room temperature in a drop consisting of 50% protein solution and 50% reservoir [2.52 M ammonium phosphate dibasic, 0.09 M Tris at pH 8.5, 3% (w/v) D-sorbitol]. All crystals were mounted using nylon loops and flash frozen in liquid nitrogen without additional cryoprotectant.

X-ray crystallography

All data were collected at cryo-temperature (100 K). The PfCopC D83A data were collected at the P11 beamline at PETRA III (DESY) [26] using a wavelength of 1.0332 Å and a Pilatus 6M detector. The PfCopC E27A and D83N data were collected at the ID23-2 beamline of ESRF [27] using a wavelength of 0.87313 Å and a Pilatus3 X 2M detector. The collected data were processed using the XDS program package [28]. Structure determination and refinement were performed using the CCP4 suite [29]. The structures were phased by molecular replacement in MOLREP [30] using the PDB code 6NFQ as the search model. The structures were refined using REFMAC5 [31] and manually rebuilt in WINCOOT [32]. A composite omit map was calculated for mutant D83A in CCP4 by omitting 20 regions of the final model, and used to make Fig. S5.

EPR spectroscopy

EPR spectra were obtained from samples submerged in liquid nitrogen (77 K) using a Bruker EMX spectrometer (Billerica, MA, USA) operating at 9.45 GHz, with a microwave power of 10.0 mW and modulation amplitude of 10 Gauss. The samples were copper loaded to completion by overnight incubation in two-times excess $CuCl_2$, and then buffer exchanged on 3 kDa Vivaspin 500 concentrators by 5 successive dilutions with 20 mM MES buffer at pH 6.6 with 40 mM NaCl. The final protein concentration was 0.5 mM. After spectra acquisition, the samples were removed from the instrument and thawed, and 10 mM ascorbic acid was added, and another series of spectra were acquired. Spectrum simulations were carried out in the MATLAB plugin EasySpin [33] using a spin system of 3 equal ^{14}N nuclei coupled to one copper atom.

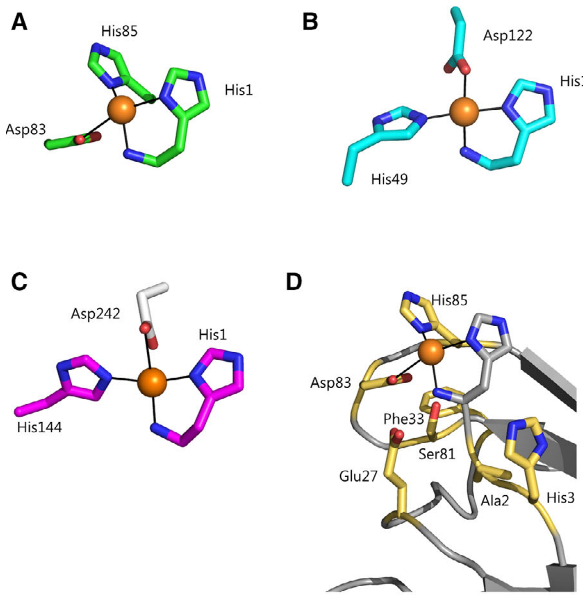
Results

Mutants designed to model His-braces found in nature

Figure 1 shows the His-brace of PfCopC, X325 and fusolin together with the conserved amino acids in PfCopC chosen for mutation (Fig. 1D). A conserved Glu27 (E27) occupies an interesting position in the second sphere of PfCopC [12,13]. This residue is located 3 Å from the N terminus, which corresponds to the length of a hydrogen bond. It was hypothesized that this residue was essential for the rigidity of the His-brace and that its removal would introduce flexibility around the copper-binding site. Five amino acid mutations were made at this position. The hydrogen-bonding properties were changed (Asn, Asp, Ser, Gln) or completely removed (Ala). In PfCopC, the Asp83 (D83) residue directly coordinates copper through its carboxylic group. This position was also mutated to remove the carboxylate coordinating the copper (Asn, Ala) or to introduce a sulphur ligand (Cys) that could potentially stabilize a softer Cu(I) redox state. A His mutant to replace D83 was constructed to mimic the Cu_B site of particulate mono-oxygenases. Furthermore, the removal of D83 could generate a free equatorial position on the copper atom, which is the site where the oxygenic co-substrate binds in LPMOs during the catalytic cycle of these enzymes [6,34]. We speculate that swapping the positions of His85 and Asp83 (D83H-H85D) in PfCopC would result in a good X325 His-brace mimic. However, this mimics the X325 position of the His residues but not necessarily the tautomeric form of LPMOs. A mutant was generated at the position of His3, which was initially suggested to be involved in copper binding. A mutation on Ala2, which was changed into a Pro to convert PfCopC from a Type B to a Type A [12]. Furthermore, two highly conserved residues, Ser81 and Phe33, were removed (Ala) [14]. The rationales for the mutations are summarized in Fig. 1E. In total, 15 amino acid mutants were generated from PfCopC using uracil excision cloning (see Methods). Proteins were expressed heterologously in the *Escherichia coli* periplasm, followed by purification using conventional column chromatography [22]. The effects of the introduced mutants were investigated by measuring ascorbate turnover and copper-binding properties. Such an assay has been used to investigate redox properties of Aβ16 peptides involved in Alzheimer's disease and has been adapted for His-brace proteins [14,24].

Ascorbate oxidation by PfCopC and mutants

PfCopC is not reduced by ascorbate and cannot oxidize ascorbate [14]. To investigate potential changes in their oxidizing capacity, the ability of all the mutants of PfCopC to oxidize ascorbate was tested after first equilibrating 1 μM protein with a range of 0–1.5 equivalents of $CuCl_2$ overnight (Fig. S1). Catalase and the Cu(I)-specific chelator bicinchoninic acid (BCA) were used to discriminate between redox activity by aqueous



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Positon	Variation	Changes	Why	Refs
Ala 2	Pro	Introduces steric constraint	Pro is conserved in type A CopC	[22]
His3	Phe	Removes a nitrogen ligand	Proposed to be a copper ligand	[22]
	Ala	Removes π-system and a nitrogen ligand		
Glu27	Ala[x]	Removes hydrogen bonding carboxylate	Highly conserved and in hydrogen bonding distance of the N-terminal	[22] [23] [24]
	Asp	Lengthens the distance to His1		
	Gly	Removes steric constraint and carboxylate		
	Gln	Changes hydrogen bonding donor to acceptor		
	Ser	Removes steric constraint by carbonyl		
Phe33	Ala	Removes π-system	Highly conserved and surface exposed	[24]
Ser81	Ala	Removes a hydrogen bond donor	Highly conserved	[24]
Asp83	Ala[x]	Removes carboxylate	Directly coordinating residue and strictly conserved	[23] [24]
	Cys	Introduces a soft metal ligand		
	His	Mimics the histidine brace of pMMOs		
	Asn[x]	Changes hydrogen bonding donor to acceptor		
Asp 83 & His85	Asp/His	X325 and fusolin mimic		[10] [26]

Fig. 1. Mutants expressed and purified in this study. (A) His-brace of PfCopC (green) (PDB: 6NFQ), (B) X325 (cyan) (PDB: 6IBJ), (C) fusolin (magenta) (PDB: 4X27), with an Asp side chain (white) from a neighbouring molecule capping the His-brace. (D) Mutant positions (yellow) generated in this study mapped onto the PfCopC crystal structure (PDB: 6NFQ). (E) Mutants generated in this study. [x] indicates mutants for which X-ray structures were determined in this study. Drawn in PYMOL 2.3.1.

copper and Cu(II)-PfCopC. Data for all mutants equilibrated with 0.5 equivalents of copper are shown in Fig. S2. As expected, some mutants of PfCopC were indistinguishable from the wild-type (WT) protein and did not oxidize ascorbate. These are H3F, H3A and F33A. The S81A mutant with 0.5 equivalents of copper does not oxidize ascorbate but the copper titration profile indicates that copper is unbound when loading in an excess of 0.5 equivalents. However, changes in the amino acid side chains at the positions of the A2, E27 and D83 residues resulted in ascorbate oxidation.

All mutants on E27 showed similar behaviour with respect to ascorbate oxidation (Fig. S1). The progression curve of the alanine mutant E27A is shown in Fig. 2 as an example. Mutants on this position showed a slow linear oxidation rate of ascorbate, and 41% of the ascorbate was oxidized within the time frame of the experiment. The addition of catalase had no effect on the observed oxidation rate. Furthermore, the addition of BCA did not eliminate the oxidation of ascorbate or change the rate (Fig. 2A,D). These observations indicate that copper remains associated with E27A.

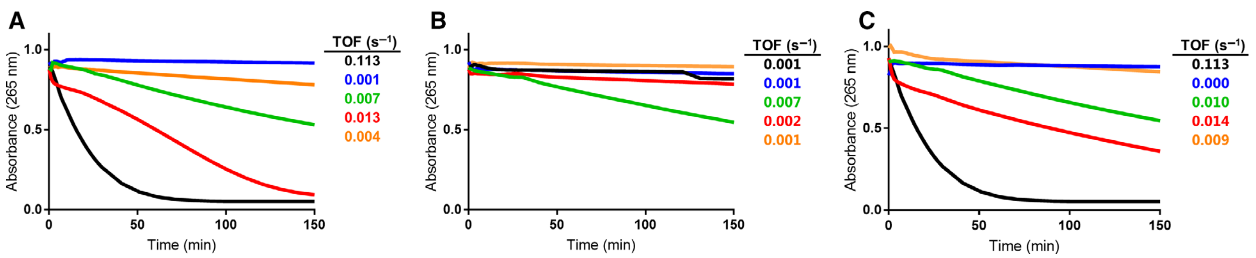

Fig. 2. Ascorbate oxidation progress curves for PfCopC and mutants (A) without scavengers, (B) with µM BCA, or (C) with 4 nm catalase. The conditions are 1 µM PfCopC (blue), E27A (green), D83A (red), D83H-H85D (orange) equilibrated with 0.5 µM CuCl2 in 20 mM MES buffer pH 6.6 and 100 µM ascorbic acid. The samples were incubated at 26 °C and 0.5 µM CuCl2 (black) was included as a control. The TOF of H_2O_2 for each condition is listed next to the graphs.

Substituting the carboxylate side chain of D83 with Ala, Cys, His or Asn residues resulted in ascorbate oxidation (Fig. S1). D83A showed a sigmoidal progress curve for ascorbate oxidation (Fig. 2A). Inclusion of BCA completely quenched oxidation (Fig. 2B), and the addition of catalase led to a linear oxidation curve (Fig. 2C). Complete oxidation of ascorbate was observed after 150 min. Ascorbate turnover frequencies were calculated for the mutants under the different conditions and presented as inserts in Fig. 2. The D83H-H85D mutant led to very slow oxidation of ascorbate (Fig. 2A). Furthermore, the oxidation of ascorbate was unaffected by catalase, but was quenched in the presence of BCA. When performing the same assay at a higher copper concentration, the oxidation rates increased, but the overall pattern was conserved (Fig. S3).

H_2O_2 production by PfCopC mutants

In order to characterize the ascorbate oxidation further, the generation of H_2O_2 was monitored using the Amplex® UltraRed assay. All the Cu–protein complexes except Cu-PfCopC produced H_2O_2 (Fig. 3A). The rate of H_2O_2 generation by Cu-E27A and Cu-D83H-H85D could be approximated by linear functions. In contrast, Cu-D83A generated H_2O_2 at a rate similar to aqueous copper. In the presence of BCA, only Cu-E27A produced measurable amounts of H_2O_2 (Fig. 3B) and the rate of H_2O_2 production was almost the same as without BCA. These observations indicate that copper remains bound to E27A, while being poorly bound by D83H-H85D and especially D83A.

UV-Vis and X-band EPR spectroscopy measurements reveal that Cu(II)-PfCopC mutants are not reduced by ascorbate

We then proceeded to investigate whether the PfCopC mutants could be reduced by ascorbate, by monitoring the absorbance in the blue wavelength region around 600 nm. Absorbance in this region is due to the *d-d* transition in Cu(II) and is the hallmark of blue copper proteins. Maximum absorbance in the range of 580–585 nm for Cu(II)–PfCopC complexes have been reported [12]. To measure reduction and not reoxidation, the absorbance was measured under anoxic conditions. As expected, PfCopC showed stable absorbance at 585 nm in the presence of 2 mM ascorbate, showing that copper is not reduced (Fig. S4). E27A was similarly stable, but this mutant exhibited a maximum absorbance at 610 nm. D83A did not exhibit any blue absorbance, consistent with a lack of copper-binding. D83H-H85D exhibited absorbance at 650 nm, which is consistent with *d-d* transitions. This is in contrast to the other findings in this study (shown below) where this mutant was found not to bind Cu(II) (Fig. 4, Table 1). EPR measurements confirmed that E27A was not reduced during 2 h of incubation. Furthermore, it was confirmed that D83A does not bind copper strongly, as no hyperfine splitting was observed (Fig. S4).

Isothermal titration calorimetry shows altered copper affinity of PfCopC amino acid mutants

Isothermal titration calorimetry was carried out to determine the affinity of these PfCopC mutants for both Cu(II) and Cu(I). Experiments were performed under both ambient and anoxic conditions. The binding constants and energies are given in Table S1. PfCopC has been shown to have a subfemtomolar affinity for Cu(II) under ambient conditions [12], while the affinity for Cu(I) has previously been determined to be 1.1×10^{-5} M [14]. The experiments here were conducted under the same conditions as in the latter study, to allow for comparison.

Apo E27A had a high affinity for Cu(II), with an apparent K_d of 1.1×10^{-8} M (Fig. 4A). This is

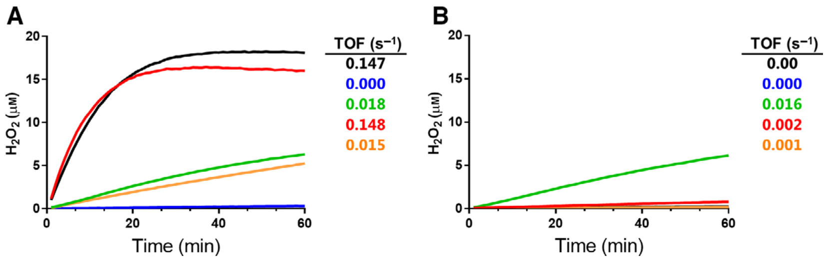

Fig. 3. Production of H_2O_2 in 100 μM ascorbate (A) without and (B) with 20 μM BCA. The samples were 0.5 μM Cu-PfCopC (blue), E27A (green), D83A (red), D83H-H85D (orange), and CuCl2 (black) in 20 mM MES buffer pH 6.6 with 50 μM Amplex® UltraRed assay reagent and 0.1 $U{\cdot}L^{-1}$ horseradish peroxidase. The samples were incubated at 26 °C and the fluorescence at 590 nm read once a minute. The TOF of H_2O_2 production for each condition is listed next to the graphs.

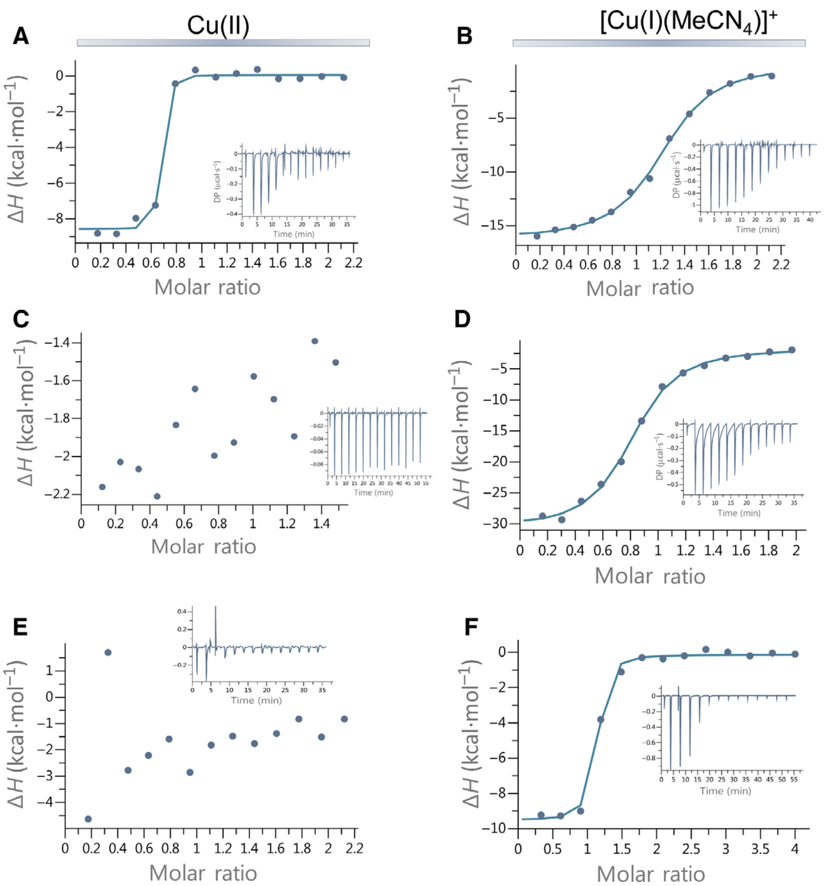

Fig. 4. ITC results of 25 µM *apo* proteins with CuCl2 (under ambient conditions) and [Cu(I)(MeCN4)]+(under anoxic conditions) in 40 mM MES with 150 mM NaCl at pH 6.6 and 25 °C. (A) E27A (ambient) shows strong Cu(II)-binding, with a suggested *Kd* of 1.1 × 10^{-8} M. (B) E27A (anoxic) shows weaker but specific binding of Cu(I) with a suggested *Kd* of 1.2 × 10^{-6} M. (C) D83A (ambient) shows no binding of Cu(II). (D) D83A (anoxic) shows binding of Cu(I) with a suggested *Kd* of 4.9 × 10^{-7} M. (E) D83H-H85D (ambient) shows no Cu(II)-binding. (F) D83H-H85D (anoxic) shows strong binding of Cu(I) with a suggested *Kd* of 1.6 × 10^{-6} M.

Table 1. Inflection points (°C) of PfCopC WT and mutants measured using DSF at protein concentrations of 0.5 mg·mL^{-1} in 40 mM MES at pH 6.6 and 150 mM NaCl. Values given are the mean of triplicate experiments ± SD, for conditions with and without 1.2 molar equivalents of copper acetate.

		With copper (°C)	Without copper (°C)
Stabilized by Cu(II)	WT	81.7 ± 0.1	60.5 ± 0.2
	E27Q	**81.3 ± 0.2**	**58.4 ± 0.1**
	A2P	**78.7 ± 0.2**	**61.9 ± 0.1**
	E27A	**75.0 ± 0.2**	**57.2 ± 0.1**
	S81A	**67.6 ± 0.2**	**59.0 ± 0.1**
	D83N	**58.0 ± 0.1**	**62.1 ± 0.2**
	D83A	58.5 ± 0.1	62.0 ± 0.2
	D83H-H85D	56.8 ± 0.0	61.4 ± 0.2

Mutants that are stabilized by Cu(II) are marked in bold.

significantly lower than the femtomolar affinity of PfCopC, but still in the low nanomolar range. The apparent affinity for Cu(I) was 1.22 × 10^{-6} M (Fig. 4B), which is slightly higher than the Cu(I) affinity reported for PfCopC [14].

Apo D83A was clearly unable to bind Cu(II) (Fig. 4C). This is interesting since it has been demonstrated that an ATCUN motif could serve as an alternative binding site when the histidines in the copper site are mutated [3,12]. Surprisingly, D83A shows an affinity for Cu(I) of 4.9 × 10^{-7} M (Fig. 4D). However, titration generates a large amount of heat and is consequently difficult to fit (Table S1). These findings suggest that D83A has improved affinity for Cu(I), but the binding site could undergo substantial reconfiguration, as indicated by the high enthalpy (Table S1).

As in the case of D83A, D83H-H85D does not bind Cu(II) (Fig. 4E), but a high affinity for Cu(I) of 1.7 × 10^{-7} M (Fig. 4F) was measured. Notably, the results of this titration were more similar to those obtained for E27A, which suggests that D83A and D83H-H85D have different modes of binding.

Differential scanning fluorimetry shows different levels of structural stabilization by Cu(II)

The thermal unfolding of the PfCopC and mutants thereof was probed using differential scanning fluorimetry (DSF). Adding Cu(II) led to differences in the inflection points as listed in Table 1. Interestingly, E27Q is stabilized more by Cu(II) than E27A, as evident by the fact that the inflection point is closer to that of the WT. The introduction of proline in A2P leads to a small increase in thermal stability of 1.4 °C without copper and a small decrease of 3 °C with copper. The removal of S81 reduced the inflection point by 14 °C. In the crystal structure, S81 is located 3 Å from D83 and could be important for His-brace stability. Surprisingly, a significant destabilizing effect was observed when adding copper to D83A, D83N and D83H-H85D, indicating that Cu(II) cannot bind to the protein when the D83 carboxylate is absent.

X-ray structures of PfCopC mutants support the biochemical observations

The structure of E27A was determined in the $C222_1$ space group to a maximum resolution of 1.54 Å (Fig. 5). A full occupancy copper in the His-brace was modelled in the single molecule found in the asymmetric unit. The anomalous difference Fourier map confirms that E27A binds copper (Fig. S6). Furthermore, the His-brace was well ordered, and coordination lengths and positions in regards to the copper were very similar to those observed for PfCopC [13]. The disordered C-terminal glutamine of a symmetry-related molecule could be modelled partly in the space occupied by Glu27 in the WT (Fig. 5B), but is not believed to be of relevance for the ability of the protein to bind copper in solution.

The structure of D83A was solved in the $I4_1$ space group to a maximum resolution of 2.15 Å. The asymmetric unit contained two molecules with significant dissimilarities at their His-braces. An artificial binding site for two coppers was formed by crystal contacts between chain A and its symmetry-related chain (Fig. S7). Each copper ion coordinates to the N-terminal nitrogen and δnitrogen of His1, a sulphate ion and a water molecule, in addition to εnitrogen of the symmetry-related His3 residue (Fig. 5C). No copper was found at the His-brace for chain B in the asymmetric unit (Fig. S7). The His-brace is itself rather disordered, and His1, His3 and His85 exhibited high B-factors of 109.9, 92.9 and 104.8 Å^2, respectively, higher than the average value for chain B of 78.8 Å^2. A composite omit map at these histidines is shown in Fig. S5. The D83N structure was solved in the same space group as D83A, to a maximum resolution of 2.3 Å. The same structural features as described above were also seen for D83N (Fig. S7). Data collection and refinement statistics are summarized in Table 2.

Discussion

In this study, we have characterized and manipulated the intriguing copper His-brace motif using PfCopC as

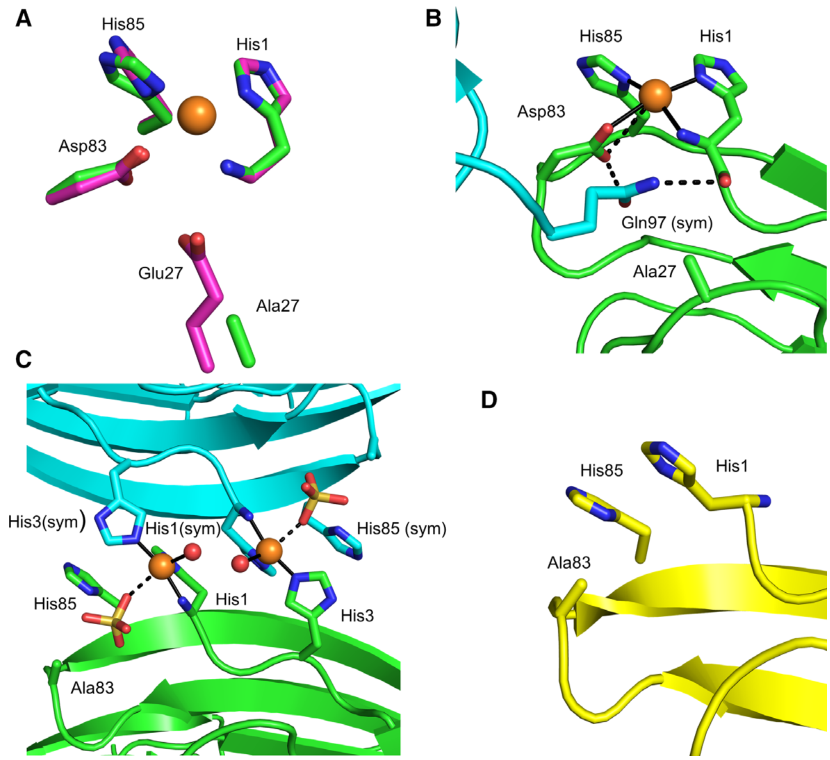

Fig. 5. Results of X-ray crystallography investigations of the D83A and E27A mutants. (A) The PfCopC structure (magenta) (PDB: 6NFQ) is superimposed on the His-brace of E27A (green). (B) The disordered crystal contact of the symmetry-related C-terminal glutamine (cyan) interacting with His1 and Asp83 in E27A. (C) The artificial copper-binding site created by the symmetry-related chains in the structure of D83A and a sulphate from crystallization conditions. (D) His-brace region of D83A chain B, showing that copper is not present at the site. The site is disordered and exhibits higher B-factors than average.

Table 2. Data collection, processing and refinement statistics. Values in parentheses are for the outer shell. The Ramachandran plot was generated using PROCHECK [43] in the CCP4 suite.

	PfCopC D83A	PfCopC E27A	PfCopC D83N
PDB ID	7BK6	7BK5	7BK7
Beamline	P11 DESY	ID23-2 ESRF	ID23-2 ESRF
Wavelength (Å)	1.0332	0.87313	0.87313
Space group	$I4_1$	$C222_1$	$I4_1$
a, *b*, *c* (Å)	80.07, 80.07, 89.39	27.00, 76.91, 81.95	80.30, 80.30, 89.89
Alpha, beta, gamma (°)	90, 90, 90	90.0, 90.0, 90.0	90, 90, 90
Resolution range (Å)	44.7–2.15 (2.21–2.15)	50–1.54 (1.58–1.54)	50–2.30 (2.36–2.30)
Completeness (%)	100 (99.7)	99.9 (100)	99.8 (99.9)
R(means) (%)	7.5 (212.7)	10.1 (175.3)	10.2 (196.2)
CC1/2 (%)	100 (54.4)	99.8 (52.5)	99.8 (45.7)
Signal to noise ratio $\langle I/\sigma(I) \rangle$	18.69 (1.07)	10.11 (1.27)	10.52 (1.31)
Total no. of reflections	210 027 (14 738)	133 453 (9867)	140 684 (10 986)
No. of unique reflections	15 379 (1136)	13 059 (951)	12 725 (954)
R(work) (%)	21.47	20.55	20.82
R(free) (%)	26.34	23.39	26.76
No. of non-H atoms			
Protein	1433	756	1490
Ions	9	10	9
Water	87	93	75
Total	1529	859	1574
Average B-factor (Å^2)			
Protein (A/B)	66.6/78.5	30.1	70.0/81.6
Metal ion	51.8	53.5	53.9
Water	68.5	39.1	67.9
RMSD bond length (Å)	0.0077	0.0101	0.0070
RMSD angles (°)	1.644	1.748	1.532
Ramachandran plot			
Most favoured (%)	92.5	96.8	93.0
Allowed (%)	5.3	3.2	5.4

a model. Five mutants of PfCopC were produced at position 27, which removed a charge stabilizing effect on the copper chelating N-terminal. All five mutants showed very slow and copper-dependent oxidation of ascorbate and generation of H_2O_2. Both effects were unchanged by scavengers, which is a clear indication that the reactivity is not simply a result of un-ligated (free) copper in the solution. This conclusion was supported by ITC data, showing that *apo* E27A binds both Cu(II) and Cu(I) with high affinity. The crystal structure of E27A shows a His-brace similar to PfCopC, but DSF experiments indicate that copper does not stabilize the mutant to the same extent as the WT. Removing a negative charge from the copper-binding environment will naturally change the affinity for the positively charged copper ions. Furthermore, the mutation also increases the flexibility of the His-brace, as seen by the loss of signal from superhyperfine couplings in the EPR spectrum. The copper is not reduced, within a 2-h timeframe, as shown by UV-Vis and EPR spectroscopy. These spectroscopic findings are contradictory to the copper-dependent slow oxidation ($0.007\ s^{-1}$) of ascorbate by E27A.

Asp83 was replaced by four different amino acid residues to give a series of PfCopC mutants at the equatorial copper ligand. Surprisingly, removal of the side chain (D83A) completely abolished Cu(II) affinity, as shown by ITC, thermal shift assays, UV-Vis and EPR spectroscopy measurements. The crystal structure shows that the His-brace is disordered, and unable to bind copper. Copper was found in the structure, but this arose from an artificial binding site created between symmetry-related molecules, which could not exist in solution. Perhaps even more surprisingly, the same was observed for the D83N substitution, which could seem more conservative at first glance. The D83 carboxylate is clearly indispensable for the copper chelating properties of PfCopC. The removal of a negatively charged carboxylate could be the reason for the observed high affinity for Cu(I). Furthermore, the ITC thermogram of Cu(I) titration to *apo*-D83A indicates major reorganization of the polypeptide chain over a timespan of approximately 3 min.

The D83H-H85D mutant mimics the *trans*-position of the histidines in the X325 His-brace and was expected to retain high affinity for Cu(II). However, the results of ITC clearly showed that this was not the case, and the mutant had a relatively high affinity for Cu(I) instead. Under ambient conditions, this resulted in very slow oxidation of ascorbate and slow generation of H_2O_2 by copper leaking into the solution. Contrary to the other experiments, under anoxic conditions, D83H-H85D exhibited absorbance indicative of an intact Cu(II)–protein complex with maximum absorbance at 650 nm. This feature disappeared following treatment with ascorbate. This is puzzling as the absorbance maximum does not match the absorbance associated with PfCopC (580–585 nm), or the alternative ATCUN copper-binding mode described previously (525 nm) [12,35]. For comparison, Cu(II)–LPMO complexes have been shown to have absorbance maximums from 615 to 650 nm [18]. The observed red-shift and peak broadening between PfCopC and D83H-H85D indicates a lower energy excited state. We speculate that this is due to stabilization of the planar coordination and increased Jahn-Teller peak splitting. However, this broad peak was not observed at ambient condition, and unfortunately, a crystal structure has yet to be obtained.

The His-brace PfCopC shares a number of similarities to other well-studied copper-binding proteins. The three first amino acids are His-Ala-His, which is a His-X-His motif, but since it is at the N-terminal it is also an ATCUN site. ATCUN is a tripeptide motif where the ligands to the metal are provided by the N terminus and a histidine sidechain in the third position. Additional nitrogen ligands are provided by the peptide backbone [36]. The ATCUN motif has a high affinity for Cu(II) but cannot be reduced as the coordination sphere is not compatible with the electronic configuration of Cu(I). However, the ATCUN site has been shown to be reduced by ascorbate when in sequence proximity to the *Bis*-histidine motif [37,38]. Our data do not support the notion that the ATCUN motif could be an alternative copper-binding site when Asp83 is mutated.

For D83A, we propose a Cu(I) site with characteristics of both His-X-His and the *Bis*-histidine site of histatin. His-Gly-His binds Cu(I) in which the imidazoles are in the *trans*-position and oxidation of the Cu(I)->Cu(II) was only observed when bound through εN [19]. We propose that His1 and His3 are in the *trans*-position and both bind *via* the δN and that His85 donates additional nitrogen. Furthermore, the carbonyl between Ala2 and His3 could be involved (Fig. 6). This could explain the high enthalpy of the binding reaction observed in ITC. Similarly to *Bis*-histidine sites, this site has a strict Cu(I)-binding mode. However, such a structure would entail considerable rearrangements from the crystal structure observed in the absence of Cu(I). This would explain why copper is essentially aqueous in our ascorbate and Amplex Red experiments.

Cu(I) clearly binds to D83H-H85D with high affinity but, as indicated by our ascorbate oxidation and Amplex Red experiments, its affinity is lower than that for BCA. In the presence of BCA, copper leaks or is extracted from the complex. Less heat was generated during titration, than with D83A, which could imply that less reorganization is taking place. Furthermore, we observed no stabilizing effect of Cu(II), meaning that Cu(I) could be bound in a linear fashion and with fewer participating ligands.

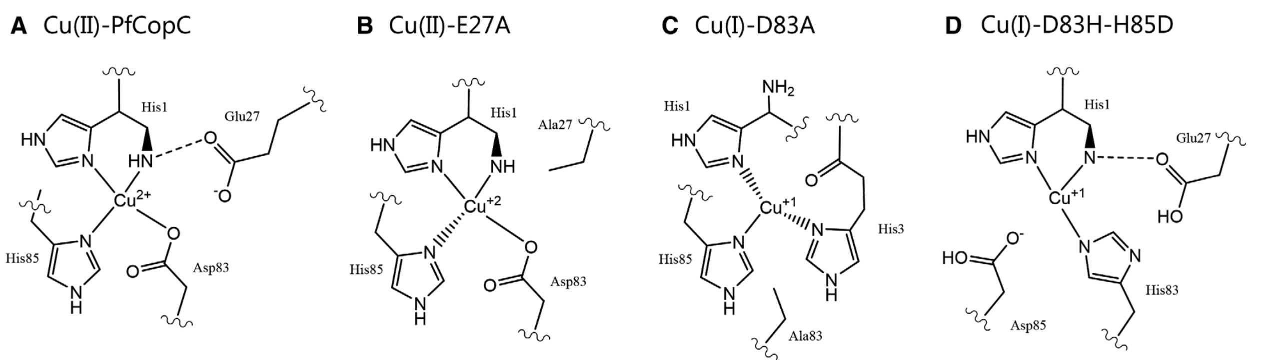

Fig. 6. Schematic models of Cu motifs in PfCopC and mutants thereof. (A) PfCopC (PDB: 6NFQ) (B) The Cu(II)-E27A is similar to the WT, as seen in the crystal structure (PDB: 7BK6, this work). (C) Possible structure of Cu(I)-D83A based on [37,38]. In the proposed structure, His1 and His3 anchor Cu(I) in a linear fashion with an additional ligand from His85 explaining the increased enthalpy of binding observed by ITC. (D) Proposed structure Cu(I)-D83H-H85D. Due the observed red shift, a planar mode is proposed.

In LPMOs, the coordinating imidazole side chains are in a *trans*-position and seem to have a tautomeric preference with the δN His1 and the εN from the internal histidine [18]. Creating mimics more closely associated with native LPMO features has been suggested to give a more accurate His-brace mimic [39]. In PfCopC, the imidazoles are in the *cis*-position and the δN is donated for both imidazolyl ligands. As shown by model His-Gly-His peptides, the imidazole tautomer is important for the redox behaviour and oxidative chemistry of the copper [19]. Interestingly, δN complexes did not show redox behaviour while εN complexes did. This mimics the behaviour of PfCopC and LPMOs and is likely important for determination of activity. However, LPMOs have a π-electron system in the axial position below the copper (Tyr or Phe) [6] which is absent in PfCopC and likely to be important for catalysis [40]. In this study, we have shown that simply mimicking the *trans*-position of the histidines or removing the carboxylate at the equatorial position is insufficient to recreate a redox-active copper site. This is in agreement with other studies of the importance of copper coordinating and second-sphere amino acid residues for LPMO activity [18,40–42].

Conclusions

Using PfCopC as a model system, we were able to gain new insights into the importance of residues directly associated with and around the His-brace-bound copper. Glu27 was found to be important for the strong binding of copper, as its removal led to the oxidation of ascorbate and the generation of H_2O_2. E27A still binds both Cu(II) and Cu(I) tightly, but is unable to redox cycle between the two. This is likely due to limited flexibility and inadequate second-sphere composition. The carboxylate donated by Asp83 is crucial for copper chelation, and removal of this side chain renders the protein nonfunctional. The X325 mimic D83H-H85D does not bind copper strongly, but is clearly not as detrimental to copper binding as the single mutant. Under anoxic conditions, D83H-H85D has an absorbance indicative of an intact Cu(II)–protein complex. This feature disappears after the addition of ascorbate. Interestingly, the mimic is able to bind Cu(I). Therefore, copper could remain associated with D83H-H85D, and initiation of the redox cycle could be possible under these specific conditions. As shown here, the positions of the coordinating atoms and the second-sphere around them play a crucial role in tuning the function.

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Author contributions

CH-R. and ABB expressed the CopC mutants; JØI purified and characterized CopC and performed the titrations and UV-Vis measurements. JØI and SB designed and performed the ascorbate assays, and SB performed the EPR measurements and subsequent analysis. SJM performed the crystallization, DSF measurements and structural analysis. LLL directed the structural studies and carried out structural analysis, and KSJ directed the overall study. All authors contributed to the preparation of the manuscript.

Data accessibility

The structural data that support these findings are openly available in the wwPDB at https://doi.org/10.2210/pdb<7BK5>/pdb, https://doi.org/10.2210/pdb<7BK6>/pdb, and https://doi.org/10.2210/pdb<7BK7>/pdb.

References

1 Solomon EI, Heppner DE, Johnston EM, Ginsbach JW, Cirera J, Qayyum M, Kieber-Emmons MT, Kjaergaard CH, Hadt RG and Tian LI (2014) Copper active sites in biology. *Chem Rev* **114**, 3659–3853.

2 Holm RH, Kennepohl P and Solomon EI (1996) Structural and functional aspects of metal sites in biology. *Chem Rev* **96**, 2239–2314.

3 Rubino JT and Franz KJ (2012) Coordination chemistry of copper proteins: how nature handles a toxic cargo for essential function. *J Inorg Biochem* **107**, 129–143.
4 Quinlan RJ, Sweeney MD, Lo Leggio L, Otten H, Poulsen J-CN, Johansen KS, Krogh K, Jorgensen CI, Tovborg M, Anthonsen A *et al.* (2011) Insights into the oxidative degradation of cellulose by a copper metalloenzyme that exploits biomass components. *Proc Natl Acad Sci USA* **108**, 15079–15084.
5 Kjaergaard CH, Qayyum MF, Wong SD, Xu F, Hemsworth GR, Walton DJ, Young NA, Davies GJ, Walton PH, Johansen KS *et al.* (2014) Spectroscopic and computational insight into the activation of O_2 by the mononuclear Cu center in polysaccharide monooxygenases. *Proc Natl Acad Sci USA* **111**, 8797–8802.
6 Ciano L, Davies GJ, Tolman WB and Walton PH (2018) Bracing copper for the catalytic oxidation of C-H bonds. *Nat Catal* **1**, 571–577.
7 Ross MO, MacMillan F, Wang J, Nisthal A, Lawton TJ, Olafson BD, Mayo SL, Rosenzweig AC and Hoffman BM (2019) Particulate methane monooxygenase contains only mononuclear copper centers. *Science* **364**, 566–570.
8 Ro SY, Schachner LF, Koo CW, Purohit R, Remis JP, Kenney GE, Liauw BW, Thomas PM, Patrie SM, Kelleher NL *et al.* (2019) Native top-down mass spectrometry provides insights into the copper centers of membrane-bound methane monooxygenase. *Nat Commun* **10**, 2675.
9 Zhang L, Koay M, Maher MJ, Xiao Z and Wedd AG (2006) Intermolecular transfer of copper ions from the CopC protein of *Pseudomonas syringae*. Crystal structures of fully loaded Cu(I)Cu(II) forms. *J Am Chem Soc* **128**, 5834–5850.
10 Lawton TJ, Kenney GE, Hurley JD and Rosenzweig AC (2016) The CopC family: structural and bioinformatic insights into a diverse group of periplasmic copper binding proteins. *Biochemistry* **55**, 2278–2290.
11 Puig S, Rees EM and Thiele DJ (2002) The ABCDs of periplasmic copper trafficking. *Structure* **10**, 1292–1295.
12 Wijekoon CJK, Young TR, Wedd AG and Xiao Z (2015) CopC protein from *Pseudomonas fluorescens* SBW25 features a conserved novel high-affinity Cu(II) binding site. *Inorg Chem* **54**, 2950–2959.
13 Udagedara SR, Wijekoon CJK, Xiao Z, Wedd AG and Maher MJ (2019) The crystal structure of the CopC protein from *Pseudomonas fluorescens* reveals amended classifications for the CopC protein family. *J Inorg Biochem* **195**, 194–200.
14 Brander S, Horvath I, Ipsen JØ, Peciulyte A, Olsson L, Hernández-Rollán C, Nørholm MHH, Mossin S, Leggio LL, Probst C *et al.* (2020) Biochemical evidence of both copper chelation and oxygenase activity at the histidine brace. *Sci Rep* **10**, 16369.
15 Osipov EM, Lilina AV, Tsallagov SI, Safonova TN, Sorokin DY, Tikhonova TV and Popov VO (2018) Structure of the flavocytochrome c sulfide dehydrogenase associated with the copper-binding protein CopC from the haloalkaliphilic sulfur-oxidizing bacterium *Thioalkalivibrio paradoxus*ARh 1. *Acta Crystallogr D Struct Biol* **74** (Pt 7), 632–642.
16 Garcia-Santamarina S, Probst C, Festa RA, Ding C, Smith AD, Conklin SE, Brander S, Kinch LN, Grishin NV, Franz KJ *et al.* (2020) A lytic polysaccharide monooxygenase-like protein functions in fungal copper import and meningitis. *Nat Chem Biol* **16**, 337–344.
17 Labourel A, Frandsen KEH, Zhang F, Brouilly N, Grisel S, Haon M, Ciano L, Ropartz D, Fanuel M, Martin F *et al.* (2020) A fungal family of lytic polysaccharide monooxygenase-like copper proteins. *Nat Chem Biol* **16**, 345–350.
18 Vu VV and Ngo ST (2018) Copper active site in polysaccharide monooxygenases. *Coord Chem Rev* **368**, 134–157.
19 Park GY, Lee JY, Himes RA, Thomas GS, Blackburn NJ and Karlin KD (2014) Copper-peptide complex structure and reactivity when found in conserved His-Xaa-His sequences. *J Am Chem Soc* **136**, 12532–12535.
20 Chiu E, Hijnen M, Bunker RD, Boudes M, Rajendran C, Aizel K, Oliéric V, Schulze-Briese C, Mitsuhashi W, Young V *et al.* (2015) Structural basis for the enhancement of virulence by viral spindles and their in vivo crystallization. *Proc Natl Acad Sci USA* **112**, 3973–3978.
21 Cavaleiro AM, Kim SH, Seppälä S, Nielsen MT and Nørholm MHH (2015) Accurate DNA assembly and genome engineering with optimized uracil excision cloning. *ACS Synth Biol* **4**, 1042–1046.
22 Hernández-Rollán C, Falkenberg KB, Rennig M, Bertelsen AB, Ipsen JØ, Brander S, Daley DO, Johansen KS and Nørholm MHH (2021) LyGo: a platform for rapid screening of lytic polysaccharide monooxygenase production. *ACS Synth Biol* **10**, 897–906.
23 Barkholt V and Jensen AL (1989) Amino acid analysis: determination of cysteine plus half-cystine in proteins after hydrochloric acid hydrolysis with a disulfide compound as additive. *Anal Biochem* **177**, 318–322.
24 Yako N, Young TR, Cottam Jones JM, Hutton CA, Wedd AG and Xiao Z (2017) Copper binding and redox chemistry of the Aβ16 peptide and its variants: insights into determinants of copper-dependent reactivity. *Metallomics* **9**, 278–291.
25 Johnson DK, Stevenson MJ, Almadidy ZA, Jenkins SE, Wilcox DE and Grossoehme NE (2015) Stabilization of Cu(I) for binding and calorimetric measurements in aqueous solution. *Dalton Trans* **44**, 16494–16505.
26 Burkhardt A, Pakendorf T, Reime B, Meyer J, Fischer P, Stübe N, Panneerselvam S, Lorbeer O, Stachnik K,

Warmer M *et al.* (2016) Status of the crystallography beamlines at PETRA III. *Eur Phys J Plus* **131**, 56.
27 Flot D, Mairs T, Giraud T, Guijarro M, Lesourd M, Rey V, van Brussel D, Morawe C, Borel C, Hignette O *et al.* (2010) The ID23-2 structural biology microfocus beamline at the ESRF. *J Synchrotron Radiat* **17**, 107–118.
28 Kabsch W (2010) XDS. *Acta Crystallogr D Struct Biol* **66** (Pt 2), 125–132.
29 Winn MD, Ballard CC, Cowtan KD, Dodson EJ, Emsley P, Evans PR, Keegan RM, Krissinel EB, Leslie AGW, McCoy A *et al.* (2011) Overview of the CCP4 suite and current developments. *Acta Crystallogr D Biol Crystallogr* **67** (Pt 4), 235–242.
30 Vagin A and Teplyakov A (1997) MOLREP: an automated program for molecular replacement. *J Appl Crystallogr* **30**, 1022–1025.
31 Murshudov GN, Vagin AA and Dodson EJ (1997) Refinement of macromolecular structures by the maximum-likelihood method. *Acta Crystallogr D Biol Crystallogr* **53** (Pt 3), 240–255.
32 Emsley P, Lohkamp B, Scott WG, Cowtan K (2010) Features and development of Coot. *Acta Crystallogr D Biol Crystallogr* **66** (Pt 4), 486–501.
33 Stoll S and Schweiger A (2006) EasySpin, a comprehensive software package for spectral simulation and analysis in EPR. *J Magn Reson* **178**, 42–55.
34 Hedegård ED and Ryde U (2018) Molecular mechanism of lytic polysaccharide monooxygenases. *Chem Sci* **9**, 3866–3880.
35 Hureau C, Eury H, Guillot R, Bijani C, Sayen S, Solari P-L, Guillon E, Faller P and Dorlet P (2011) X-ray and solution structures of Cu(II)GHK and Cu(II)DAHK complexes: influence on their redox properties. *Chemistry* **17**, 10151–10160.
36 Harford C and Sarkar B (1997) Amino terminal Cu(II)- and Ni(II)-binding (ATCUN) motif of proteins and peptides: metal binding, DNA cleavage, and other properties. *Acc Chem Res* **30**, 123–130.
37 Schwab S, Shearer J, Conklin SE, Alies B and Haas KL (2016) Sequence proximity between Cu(II) and Cu(I) binding sites of human copper transporter 1 model peptides defines reactivity with ascorbate and O_2. *J Inorg Biochem* **158**, 70–76.
38 Conklin SE, Bridgman EC, Su Q, Riggs-Gelasco P, Haas KL and Franz KJ (2017) Specific histidine residues confer histatin peptides with copper-dependent activity against *Candida albicans*. *Biochemistry* **56**, 4244–4255.
39 Fukatsu A, Morimoto Y, Sugimoto H and Itoh S (2020) Modelling a 'histidine brace' motif in mononuclear copper monooxygenases. *Chem Commun* **56**, 5123–5126.
40 Harris PV, Welner D, McFarland KC, Re E, Navarro Poulsen J-C, Brown K, Salbo R, Ding H, Vlasenko E, Merino S *et al.* (2010) Stimulation of lignocellulosic biomass hydrolysis by proteins of glycoside hydrolase family 61: structure and function of a large, enigmatic family. *Biochemistry* **49**, 3305–3316.
41 McEvoy A, Creutzberg J, Singh RK, Bjerrum MJ and Hedegård ED (2021) The role of the active site tyrosine in the mechanism of lytic polysaccharide monooxygenase. *Chem Sci* **12**, 352–362.
42 Jones SM, Transue WJ, Meier KK, Kelemen B and Solomon EI (2020) Kinetic analysis of amino acid radicals formed in H_2O_2-driven CuI LPMO reoxidation implicates dominant homolytic reactivity. *Proc Natl Acad Sci USA* **117**, 11916–11922.
43 Laskowski RA, MacArthur MW, Moss DS and Thornton JM (1993) PROCHECK: a program to check the stereochemical quality of protein structures. *J Appl Crystallogr* **26**, 283–291.

Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Fig. S1. Ascorbate turnover rates for PfCopC variants.

Fig. S2. Ascorbate turnover progress curves for PfCopC variants.

Fig. S3. Ascorbate oxidation progress curves for PfCopC and mutants without (a) and with the scavengers (b) 20 μM BCA, and (c) 4 nm catalase.

Fig. S4. UV-Vis traces of ascorbate reaction with Cu (II)-PfCopC complexes under anoxic conditions.

Fig. S5. Electron density at the disorded histidine brace of PfCopc D83A chain B.

Fig. S6. Anomalous difference Fourier maps are shown in Magenta at the His-brace of E27A (a), at the artificial copper binding sites of chain A in D83A (b) and D83N (c), and at the His-brace in chain B of D83A (d) and D83N (e) clearly showing binding at the artificial dimer site, and no binding to the monomeric protein, consistent with the solution studies. The maps are contoured at the 5.0 σ level and carved at 5 Å from the copper atoms in a, b and c and carved at 8 Å from the residues presented as sticks in d and e.

Fig. S7. (a) The artificial copper binding site created by the symmetry-related chains A in the structure of D83A and a sulphate from crystallization conditions. (a) His-brace region of D83N chain B, showing that copper is not present at the site. The site is disordered and exhibits higher B-factors than average.

Table S1. (a) ITC parameters (K_D, ΔH, ΔG and goodness of fit (χ^2) of titrations performed. (a) Ambient. (b) Anoxic. *Apo* proteins where titrated with $CuCl_2$ at ambient conditions and with $[CuI(MeCN_4)]^+$ inside an anoxic environment. Data for PfCopC K_d is taken from [22] in a) and [43] in b).